

Single Expansion Joint

The simplest form of Expansion joint of Single / Axial Bellows Construction, designed to absorb all of the movements of the pipe section in which it is installed according to Operated pressure of a device this model is manufactured butt weld connection, fixed flange, and loose flange with sleeve or without sleeve.

Design Rules

The selection of the design and manufacturing standard of the expansion joint is made according to specific requirements of the customer. Commonly used are:

- ASME SEC VIII DIV 1, Appendix 26
- ASME B31.3, Appendix X
- EJMA (Latest Edition)

"Standard" Expansion joints, which are listed in this Catalogue, are calculated according to EJMA. The design standard can be changed or adopted to specific customer and application requirements.

They are designed to withstand hydrostatic pressure with an adequate safety factor for industrial applications. All Expansion joints manufactured by **FLEXIBEL** are subjected to a Hydro test, regardless of the specific requirement of the customer.

Restraining parts are dimensioned to absorb axial pressure thrust and limit the lateral or angular movement of the bellows.

Materials

For a correct selection of materials to be used, the following should be considered:

1. Chemical compatibility of the material with the fluid to be transported.
2. Application –compatible mechanical strength (Temperature and Pressure)
3. Welding properties and ductility.
4. Material certification.

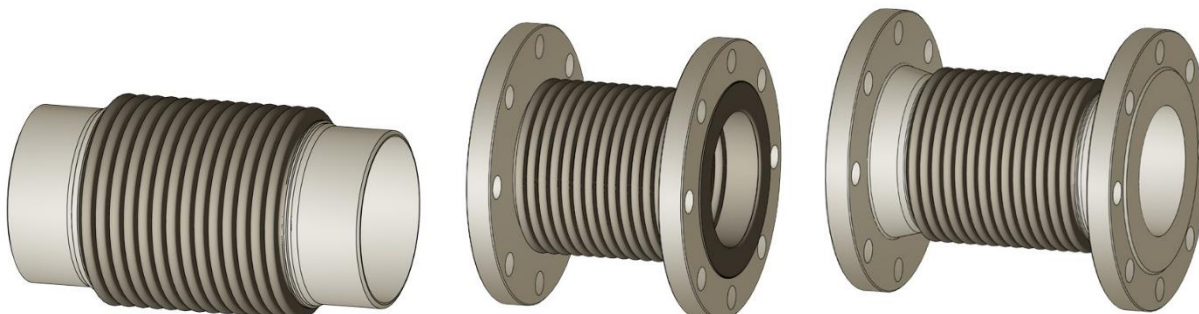
Established materials for usage in the construction of medium touched are:

Bellows: 1.4541 (AISI 321); 1.4301 (AISI 304); 1.4306 (AISI 304L); 1.4401 (AISI 316); 1.4404 (AISI 316L); 1.4571 (AISI 316Ti); 1.4828 (AISI 309); 2.4816 (Inconel 600); 2.4856 (INCONEL 625/LCF); 1.4876 (Incoloy 800/H)

Connections: ASTM A 106 Gr. B; ASTM A312 TP 304 / 316 / 321; A 240 TYP 304 / 316 / 321; ASTM A 516 GR 70.

Flanges: ASTM A 105; ASTM A 182 GR.F 304 / 316 / 321; A 240 TYP 304/316/321; ASTM A 516 GR.70.

The raw material can be changed according to the specific requirement / customer request. And also other materials are available on request.



AxWe

This type of Expansion Joint is made up of One Single bellows provided with welding Ends.

AxLf

This type of Expansion Joint is made up of One Single bellows Equipped with Floating flanges.

AxFf

This type of Expansion Joint is made up of One Single bellows Equipped with Fixed flanges.

Operating Range

Diameter : DN 100 DN 3200 (4"128").
Temperature : 700 Deg.c (1292 Deg.F).
Pressure : Max. 50 bar.(g)

Features

- Untied Metal expansion joints for General Pressure Application.
- Designed to cover the mid temperature and Pressure range for general process equipment.
- Mainly Designed for Axial Movements.
- Multilayer bellows with Low spring rate with High Cycle life.

Application

Having the ability to compensate for axial movements and with its simple and compact overall dimensions, axial expansion joints are very widely used within a range of applications.

They are especially common in long pipe runs, examples of which would include exhaust systems, ventilation and flue gas systems, district heating, steam, oil and Gas pipe systems.

Advantages / Characteristics

- Simple solution for compensation of Temperature fluctuations.
- No change in flow direction
- Compact and space saving design.
- Low cost design
- Corrosion Resistance.
- Temperature Resistance.
- Tightness,
- Pressure Resistance.
- Mechanical Strength.
- Flexibility.
- Fatigue strength.
- Freedom from Maintenance.

Therefore, they are used in the industry for:

- Compensation of thermal expansion due to temperature variation in pipes.
- Absorption of vibratory movements and acoustic vibrations.
- Compensation of piping misalignments
- Mounting adjustments.
- Reduction of forces and resulting movements in connections and anchor points.

Recommended Application

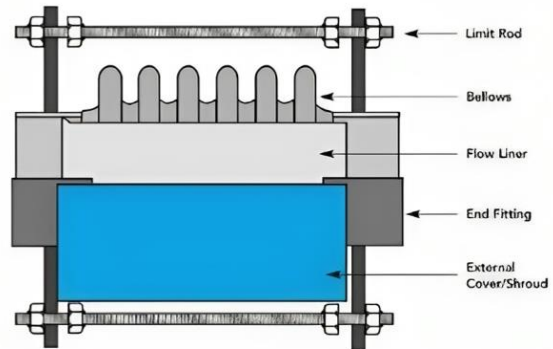
- Oil and gas industry
- Power plant technology
- Paper industry.
- Refining plant technology
- Water and waste water technology.
- Food and beverage industry.
- Ship building
- Conventional boilers
- Heat Exchangers
- Condensers
- Gas Plant technology
- Coal feed transport systems
- Water Colling systems
- Air and steam transport lines.
- Compressor / Pumping stations
- Lubrication systems

Axial Movement

Axial movement is considered as an extension or compression of the pipe system in its longitudinal axis, meaning that in the process of absorbing the movements, the overall length of the expansion joint will either extend or compress.

Axial Expansion joints which are designed to absorb large movements, can contain one or several bellows in one unit, and larger movements can also be achieved by pre-tensioning or by installing several expansion joints on the pipe section. Axial expansion joints have the ability to absorb minor lateral and angular deflections. However, we recommend the utilization should be limited to its main function, otherwise its service life may be negatively impaired.

Where there is a need for additional movement other than that of axial movements, customized design also available based on the request.



Item description

Cover – Used to protect the expansion joint.

Flow Liner / sleeve – Protects bellows from erosion damage due to abrasive media or turbulent flow.

Tie –rod – Used to retain the pressure thrust.

Limit Rod – restrict bellows movement range and prevent over compression or over extension.

Control Rod – Used to distribute movement between 2 bellows in a universal expansion Joint.

Adjustment rod – Used to adjust the length of the expansion joint prior to installation.

Pipe / Flange – Used to mount the pipe line. By using bolting / welding.

Requirements

- Strong fix points and good guides are required.
- For Larger movements multiple axial expansion joints required.
- Many fix points and guides are needed for longer pipe.

Types of Compensation

Axial

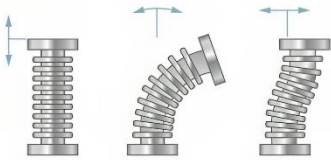
- Compression and extension of the metallic bellows
- Combination with lateral and angular deflection possible.
- Little space for installation required.
- Length adjustment with tensioners possible.
- Long straight pipes with several expansion joint need intermediate anchor points.
- Transmits pressure thrust to the adjacent pipe supports.

Angular

- Angular deflection
- Does not allow axial deflection. Pipe redirection necessary.
- Minimum two expansion joints needed.
- Absorbs pressure thrust low loads on adjacent pipe supports.

Lateral

- Lateral deflection
- Relatively simple design for low pressure.
- Does not allow axial deflection. Pipe redirection necessary.
- For high movement compensation.
- Absorbs pressure thrust. Low loads on adjacent pipe supports.



Note: Metal Expansion joints should never be subjected to torsion. They should be positioned in the piping systems in order to avoid such loads.

Installation instruction

- The bellows element shall be protected from damage. Dents, scores, arc strikes, weld spatter, and other damage can cause the joint to fail. Damaged joints should not be used.
- Align joint flange and pipe flange holes. Do not try to compensate for flange or pipe misalignment by putting any torsional, compressive, extension, or offset loads on the expansion joint. Good practice suggests that a mating flange in the piping system remain un-welded until the expansion joint has been bolted in position.
- All anchors, guides, and supports must be installed according to engineering drawings and specifications.
- Internally pressurized expansion joints are to be installed in the proper orientation with respect to direction of flow.
- Unit lengths must not be altered during installation except for the application of cold pull.
- Remove shipping restraints after installation, but before hydro testing.
- Test pressure should not exceed 1.5 times of the design pressure.
- Water free of halogens should be used for hydro testing.
- If testing medium is significantly heavier than the product to be carried in the system, care must be taken to support the additional weight.
- Paints containing low melting point metals or their compounds, particularly aluminum, lead or zinc, must not be allowed to come into contact with the bellows convolutions.
- If high frequency vibrations or turbulence or higher flow velocity are to be expected in the medium, we recommend the installation of expansion joints with internal sleeves.

CIRCULAR BELLOWS DATA SHEET

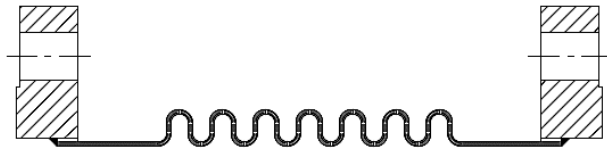
CUSTOMER :		DATE		PAGE
PROJECT :		PREPARED BY :		
APPLICABLE CODES AND STANDARDS : ASME SEC VIII / EJMA				
ITEM OR TAG NUMBER :				
QUANTITY :				
SIZE :				
ORIENTATION :				
TYPE (SINGLE / UNIVERSAL /HINGED / GIMBAL., ETC				
END CONNECTIONS	THICKNESS / FLANGE RATING / DIMENSIONS			
	MATERIAL			
PRESSURE	DESIGN			
	OPERATING			
	TEST			
TEMPERATURE	DESIGN			
	OPERATING			
	INSTALLATION			
MEDIA	FLOW MEDIA			
	FLOW VELOCITY			
	FLOW DIRECTION			
MOVEMENTS & CYCLE LIFE	INSTALLATION	AXIAL EXTENSION		
		AXIAL COMPRESSION		
		LATERAL		
		ANGULAR		
		NO OF CYCLES		
	DESIGN	AXIAL EXTENSION		
		AXIAL COMPRESSION		
		LATERAL		
		ANGULAR		
		NO OF CYCLES		
	OPERATING	AXIAL EXTENSION		
		AXIAL COMPRESSION		
		LATERAL		
		ANGULAR		
		NO OF CYCLES		
MATERIALS	BELLOWS			
	LINER			
	COVER			
	END CONNECTION			
DIMENSIONS	OVERALL LENGTH			
	MAXIMUM OUTER DIAMETER.			
	MINIMUM INTERNAL DIAMETER			
SPRING RATE	MAXIMUM AXIAL SPRING RATE			
	MAXIMUM LATERAL SPRING RATE			
	MAXIMUM ANGULAR SPRING RATE			
QUALITY REQUIREMENTS				
SPECIAL REQUIREMENTS				



Connection Option

Weld Ends

Each End Connection can be defined individually.



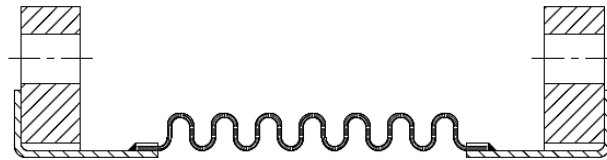
Connection Option

Fixed Flanges

Each End Connection can be defined individually.

Flange Dimensions as per Standard / Customer Specification.

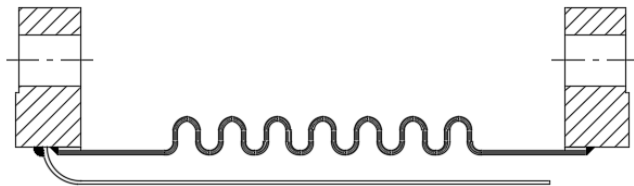
Flange Dimensions can be customized.



Connection Option

Loose liner / Swivel Flanges

- Easy installation, flange can be rotated.
- Flanges are not in contact with medium, which enables lower grade material.
- Connection with non-weldable materials possible.
- Galvanic surfaces / Coatings Possible
- Gaskets can often be eliminated.

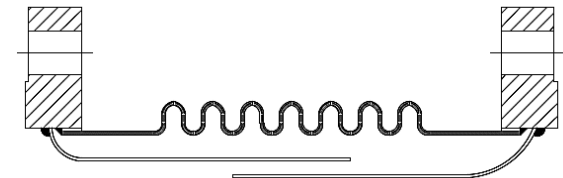


Internal Liner Option

Fixed Liner

If Necessary, inner sleeve may be used to protect the expansion joint in order to:

- Reduce the flow resistance
- Limit flow-induced vibration.
- Protect the bellows from damage from direct contact with the flowing medium and to
- Avoid deposition of solid particles in the metal bellows.

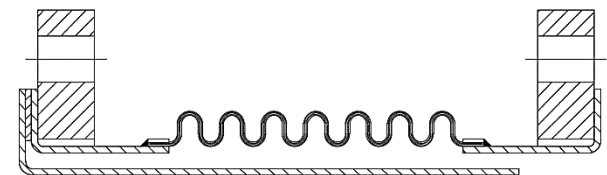


Internal Liner Option

Fixed telescopic sleeve

If Necessary, inner sleeve may be used to protect the expansion joint in order to:

- Larger Axial movements.
- When there is possibility for reverse flow, or bi-directional flow use of telescoping liner is recommended.
- Reduce the flow resistance
- Limit flow-induced vibration.
- Protect the bellows from damage from direct contact with the flowing medium and to
- Avoid deposition of solid particles in the metal bellows.



Connection Option

Loose liner / Swivel Flanges

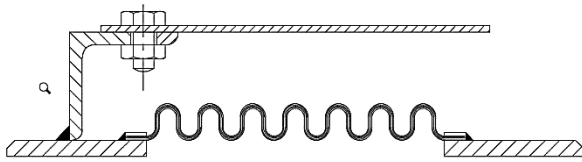
Easy installation, flange can be rotated.

Flanges are not in contact with medium, which enables lower grade material.

Connection with non-weldable materials possible.

Galvanic surfaces / Coatings Possible

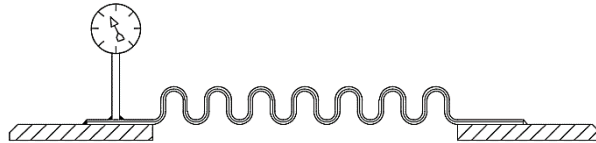
Gaskets can often be eliminated.



External Cover Option

Protective cover

Metal covers are installed on the expansion joint to protect the thin metal bellows.

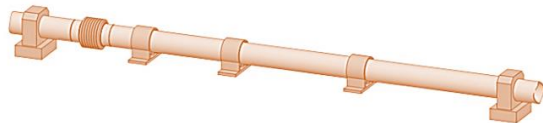


2-ply testable bellows

2-Ply Testable bellows, enables monitoring of the bellows inner / outer ply integrity.

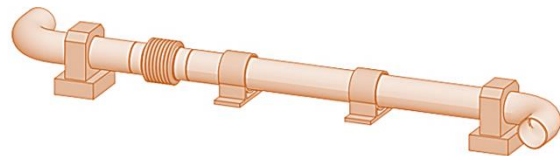
Single Bellows Applications

Type-1



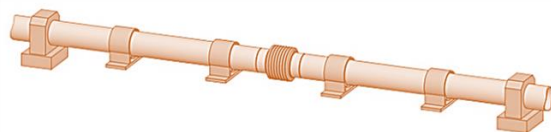
The above image shows an Expansion joint located in a straight section of piping installation between two Fixed Points.

Type-2



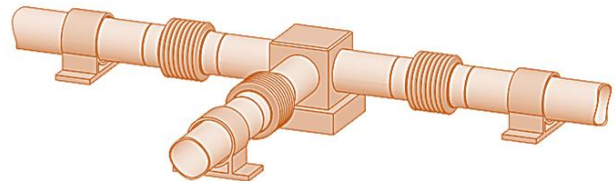
The above image shows an Expansion joint located where pipes changes directions in order to the straight section.

Type-3



The above image shows an Expansion joint located in a straight section of pipe, the axial bellows fitted in a way that they are joined together by an intermediate points.

Type-4



The above image shows an Expansion joint located at intersection where two sections of piping meet.